#Lab work. #IoT

Note: You can prepare labwork yourself as you did in the class and in the workshop. You can add the lab work as follows.

1. **Temperature and Humidity Sensing**
   - **Objective**: Interface a DHT11 or DHT22 sensor with an ESP8266/Arduino and display the temperature and humidity readings on a web page.
   - **Task**: Measure the environment's temperature and humidity every 5 seconds and display the data on a local web server.

2. Motion Detection and Alert System

- **Objective**: Use a PIR sensor to detect motion and trigger an alert using an LED or buzzer.
- **Task**: Create a system that sends a notification via email or SMS when motion is detected within a room.

3. Smart Lighting Control with LDR

- **Objective**: Interface a Light Dependent Resistor (LDR) to control an LED or a light bulb based on ambient light intensity.
- **Task**: Program a system where lights automatically turn on when the light level drops below a threshold.

4. Distance Measurement Using Ultrasonic Sensor

- **Objective**: Use an HC-SR04 ultrasonic sensor to measure the distance to an object.
- **Task**: Create a simple application that displays the measured distance on an LCD or sends the data to a server for monitoring.

5. Soil Moisture Sensor-Based Irrigation System

- **Objective**: Build an automatic irrigation system using a soil moisture sensor.
- **Task**: When the soil moisture level drops below a set threshold, turn on a water pump (actuator), and turn it off once the desired moisture level is reached.

6. Gas Detection and Monitoring

- **Objective**: Interface an MQ2/MQ135 gas sensor to detect smoke or harmful gases.
- **Task**: Create a system that sends an alert to a smartphone via Wi-Fi when gas levels exceed a safe limit.

7. Servo Motor Control via IoT

- **Objective**: Control the position of a servo motor remotely via a web interface.
- **Task**: Develop an IoT application to control the servo motor from a smartphone app or web interface.

8. Heart Rate Monitoring System

- **Objective**: Interface a pulse sensor to measure heart rate.
- **Task**: Display the heart rate data on an OLED or LCD screen and send the data to a cloud platform for real-time monitoring.

9. Environmental Monitoring with Multiple Sensors

- **Objective**: Interface multiple sensors (temperature, humidity, air quality) and send the data to an IoT platform like ThingSpeak or Blynk.
- **Task**: Use MQTT or HTTP to send data at regular intervals to the cloud, where students can visualize it on a dashboard.

10. Home Automation Using Relay and Wi-Fi Module

- **Objective**: Control home appliances using a relay module connected to a Wi-Fi-enabled microcontroller (e.g., ESP8266/ESP32).

- **Task**: Design a home automation system that allows users to control appliances like fans or lights via a mobile app or voice commands.